BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 047 · Lesson 047 · Two- and Three-dimensional Arrays in C#

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does index 0 work in C# arrays?

Q2. What is a property on a C# object?

Q3. What is the distinction between an array reference and an object instance?

Q4. 017

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I Do · Read new knowledge

Key vocabulary

dimension — An indexing axis of an array.

row — First index in 2D example.

column — Second index in 2D example.

voxel — 3D grid element.

New learning

Observe — A rectangular C# 2D array uses [row,column]. Example seats has 2 rows and 3 columns.

Explain — A 3D array uses [layer,row,column]. new int[2,3,4] holds 24 integers; its last valid index is [1,2,3].

Apply — Each dimension starts at zero; distinguish [1,2,3] from a single flat offset and never exceed a dimension's length.

Worked example

Find seats[0,2] in {{3,5,7},{9,11,13}}.

1. Trace and justify — First row, third column gives 7. The column index 2 is valid because there are three columns.

We Do · Practise with guidance

Work out final value from a different 3D array.

```
int[,,] cubes=new int[2,2,3];
cubes[1,0,2]=14;
Console.WriteLine(cubes[1,0,2]);
```

Trace and justify

Output 14: second layer, first row, third column. Six values per layer, twelve total.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Find a[1,1] in {{2,4,6},{8,10,12}} and show indexes. [4]

Question 2

A 3D array has sizes [2,3,4]. Calculate items and last index triple. [6]

Question 3

Write two nested C# loops that print all values in int[,] a of 3 rows and 2 columns, explaining order. [6]

Exit Ticket · Answer before checking

In int[,,] x=new int[2,3,4], x[2,0,0] is valid.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It accesses the first position.

Q2. A member exposing data through get/set accessors.

Q3. A reference points to the object; the object holds state.

Q4. A zero-indexed 4×3 array has how many items and what last valid (row,column)?

You Do pages · Independent questions

Q1. 10 [2]; second row (1), second column (1) [2].

Q2. 2×3×4=24 [3], last triple [1,2,3] [3].

Q3. for(int r=0;r<3;r++) [2] for(int c=0;c<2;c++) [2] Console.WriteLine(a[r,c]); [2]. Row-major traversal visits (0,0)..(2,1).

Exit Ticket · Model answer

False. First dimension indexes are only 0 and 1.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: